MIL-A-27195(USAF)
AMENDMENT 2
8 FEBRUARY 83
Superseding
AMENDMENT 1
25 April 1961

MILITARY SPECIFICATION

ACCELEROMETER, AIRCRAFT ABU-6/A

This amendment forms a part of Military Specification MIL-A-27195(USAF), dated 27 May 1959 and is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

*2.1: Delete all documents listed on page 1 and page 2 and insert the following:

"SPECIFICATIONS

Federal

PPP-B-601	-	Boxes, Wood, Cleated Plywood
PPP-B-636	-	Box, Shipping, Fiberboard

Military

MIL-P-116	-	Preservation, Methods of
DOD-D-1000	-	Drawings, Engineering and Associated Lists
MIL-C-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-6872	-	Soldering Process, General Specification for
MIL-A-8625	-	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-L-25467	-	Lighting, Integral, Red, Aircraft Instrument, General Specifications for
MIL-P-26514	-	Polyurethane Foam, Rigid or Flexible, for Packaging
MIL-L-27160	-	Lighting, White, Integral, Instrumental, General Specification for
MIL-C-83488	-	Coating, Aluminum, Ion Vapor Deposited

STANDARDS

Federal

FED-STD-595	-	Colors

*Add the following boxed paragraph to bottom of page:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6610

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Military

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-143	-	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-794	-	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	-	Environmental Test Methods
MIL-STD-831	-	Test Reports, Preparation of
MIL-STD-838	-	Lubrication of Military Equipment
MIL-STD-889	-	Dissimilar Metals
MS 33558	-	Numerals and Letters, Aircraft, Instrument Dial, Standard Form of
MS 33585	-	Pointer, Dial, Standard Design of Aircraft Instrument
MS 33638	-	Case, Instrument, Flange Mounted, Aircraft

Delete "PUBLICATIONS" and documents listed thereunder.

3.1: Delete paragraph and substitute the following:

"3.1 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.5 and 6.4)."

3.2, line 3: Delete "ANA Bulletin 143" and substitute "MIL-STD-143".

PAGE 3

*3.3.5, line 4: Delete "MS 33586" and substitute "MIL-STD-889".

PAGE 4

*Figure 1: Delete "3-7/8 ± 1/4" and substitute "3-7/8 maximum". Delete reference to "MS 33548" and substitute "MS 33638".

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*3.6, line 3: Delete "MIL-D-5028" and substitute "DOD-D-1000".

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*3.15, line 2: Delete "MIL-L-6880" and substitute, "MIL-STD-838".

*3.17.2: Delete entirely and substitute the following:

"3.17.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practical, in accordance with MIL-C-83488, Type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coatings, in lieu of MIL-C-83488, may

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be used if demonstrated to be satisfactory and is approved by the cognizant engineering activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

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*4.1,b: Delete "Preproduction" and substitute "First article".

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*4.4.8: Delete and substitute new paragraph:

"4.4.8 Low temperature operation. The accelerometer shall be subjected to the low temperature test, Method 502.1, Procedure I of MIL-STD-810."

*4.4.9: Delete and substitute new paragraph:

"4.4.9 High temperature operation. The accelerometer shall be subjected to the high temperature test, Method 501.1, Procedure I of MIL-STD-810 for a period of 4 hours. Upon completion of the test period and while at the test temperatures, the accelerometer shall be subject to and meet the individual tests."

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*4.4.12: Delete paragraph and substitute the following:

"4.4.12 Vibration endurance. The instrument shall be mounted on a suitable vibration table in its normal operating position. The accelerometer shall be vibrated with circular motion of 0.018 to 0.020 inch diameter in a plane inclined 45 degrees to the horizontal plane and the frequency of vibration shall be varied uniformly from 5 to 50 cycles per second and return once each hour for a 50-hour period. Upon completion of this test period the accelerometer shall be subjected to the tests specified in 4.4.3. The dynamic scale error shall not exceed 0.2g."

*4.4.15: Delete paragraph entirely and substitute the following:

"4.4.15 Low temperature - high altitude. The accelerometer shall be subjected to the low temperature test, Method 502.1, Procedure I of MIL-STD-810, and an altitude of 70,000 feet (1.31 inches Hg); except the temperature shall be held at the lowest operating temperature for a period of 72 hours."

*4.4.16: Delete "Procedure I of MIL-E-5272" and substitute "Method 501.1. of MIL-STD-810."

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*4.4.18: Delete paragraph and substitute new paragraph as follows:

"4.4.18 Humidity. The accelerometer shall be subjected to the humidity test in accordance with Method 507.1, Procedure I, of MIL-STD-810, except the temperature shall be 71°C (160°F) in step 2."

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*4.4.19, line 2: Delete "Procedure I of MIL-E-5272" and substitute "Method 508.1 of MIL-STD-810".

*4.4.20, line 2: Delete "Procedure I of MIL-E-5272" and substitute "Method 509.1, Procedure I, of MIL-STD-810."

*4.5 through 4.5.4: Delete the word "Preproduction" wherever referenced and substitute "First article".

*Add new paragraph as follows:

"4.6 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification."

*Delete Section 5 entirely and substitute the following:

"5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified.

5.1.1 Level A.

5.1.1.1 Cleaning. Accelerometers shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Accelerometers shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each accelerometer, aircraft ABU-6/A, shall be packaged in quantity unit packs of one each in accordance with method IC-1 of MIL-P-116. Each accelerometer shall be overpacked in PPP-B-636 carton. Carton shall be large enough to allow for application of sufficient cushioning material, MIL-P-26514, between container and bag, of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the transmitter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Accelerometers shall be clean, dry and individually packaged in a manner that will afford protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified. (See 6.2)

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5.2.1 Level A. Accelerometers packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A and B, class overseas unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, and of minimum tare and cube, consistent with the protection required.

5.2.2 Level B. Accelerometers packaged as specified in 5.1.1 shall be packed in weather resistant exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above are applicable.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

5.3 Marking of shipment. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Accelerometer, Aircraft ABU-6/A

5.3.1 Reinspection date. Reinspection date markings shall be in accordance with the instructions of the procuring activity."

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6.2: Add "e. Reinspection date".

*Add new paragraph as follows:

"6.4 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article."

*Add new paragraph as follows:

"6.5 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement STANAG 3330 and Air Std 10/4. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations."

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NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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